

Sri Lanka Institute of Information Technology



IE2034 – Analog Electronics

Final Assignment Report

Group 16

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Abstract

This report presents the design, construction, and testing of a three-stage analog signal conditioning circuit intended for use as the audio output section of a basic AM radio receiver. The circuit consists of a first-order high-pass RC filter with a cutoff frequency of 2 kHz, a single-stage common-emitter BJT amplifier using a 2N2222 transistor, and a first-order low-pass RC filter with a cutoff frequency of 12 kHz. The high-pass filter was merged with the amplifier's input biasing network by sharing an 8.2 k Ω resistor, which reduced component count and eliminated inter-stage loading. The amplifier was biased to operate from a 9V DC supply with a Q-point at $V_{CE} \approx 5.4V$ and $I_C \approx 0.64$ mA, ensuring linear operation. A 100 μ F bypass capacitor across the emitter resistor maximized AC gain at signal frequencies. The circuit was constructed on a dot board and tested using a function generator and oscilloscope at input frequencies of 5 kHz, 12 kHz, and 16 kHz with an input amplitude of 100 mVpp. Measurement results showed that the amplifier achieved a voltage gain of 62 to 64 across the passband, exceeding the required gain of 60. The high-pass filter effectively attenuated frequencies below 2 kHz, while the low-pass filter began attenuating frequencies above 15 kHz, providing useful high-frequency noise rejection. The output waveform remained clean and distortion-free with the expected 180° phase shift, confirming proper common-emitter operation. All design requirements were successfully met, and the circuit is suitable for processing weak AM radio signals into clear audio output.

Table of Contents

Abstract	2
Introduction	4
Theoretical Calculations	5
Cutoff Frequencies.....	5
Amplifier Gain	6
Design Choices for Components	10
Circuit Design & Diagram	12
Output Waveform	13
Observations and Explanations	14
Discussion.....	16

Introduction

In the field of analog electronics, raw signals from transducers or communication receivers are rarely in a form suitable for direct use. They often require conditioning, filtering to remove unwanted frequency components and amplification to raise signal strength to a usable level. This is particularly true in a basic AM radio receiver, where the demodulated audio signal is weak and contaminated with both low-frequency noise (such as power line hum and wind noise) and high-frequency artifacts (such as the residual carrier signal and high-frequency interference).

The objective of this assignment is to design, build, and test a multi-stage analog signal conditioning circuit that simulates the audio output section of an AM radio. The circuit must accept a weak, noisy audio signal and produce a cleaner, amplified output suitable for driving headphones or a small speaker.

The design is divided into three sequential stages, each with a specific function:

1. **Stage 1: High-Pass Filter** – A passive RC high-pass filter with a cutoff frequency of 2 kHz is used to attenuate low-frequency noise below the audio band of interest.
2. **Stage 2: Transistor-Based Common Emitter Amplifier** – A BJT (Bipolar Junction Transistor) in a common-emitter configuration provides a mid-band voltage gain of at least 60, amplifying the filtered signal to a suitable level.
3. **Stage 3: Low-Pass Filter** – A passive RC low-pass filter with a cutoff frequency of 12 kHz removes high-frequency noise, smoothing the signal for clear audio output.

The entire circuit is designed to operate from a 9V DC power supply, with the BJT amplifier properly biased to remain in its active region throughout operation. Following the design calculations, the circuit is constructed on a dot board (perfboard) using discrete passive components and a BJT transistor. Each stage is tested individually and then as a complete system.

This report documents the full design process, including:

- Theoretical calculations for cutoff frequencies and amplifier gain
- Justification for all component choices
- A complete circuit diagram
- Oscilloscope measurements of the signal at each stage
- Observations and explanations of signal behavior
- A discussion on the overall performance of the amplifier and the effectiveness of the filtering stages

Through this practical implementation, the assignment demonstrates how fundamental analog circuit blocks. RC filters and a BJT amplifier—can be integrated to solve a realistic engineering problem in radio audio processing.

Theoretical Calculations

Cutoff Frequencies

Stage 01: High-Pass Filter (Cutoff Frequency = 2 kHz)

The cutoff frequency for a first-order RC high-pass filter is given by:

$$f_c = \frac{1}{2\pi RC}$$

Where:

- f_c = Cutoff frequency (Hz)
- R = Resistance (Ω)
- C = Capacitance (F)

Component Selection:

To achieve $f_c = 2$ kHz, I selected standard component values:

- $C = 10$ nF = 10×10^{-9} F
- $R = 8.2$ k Ω = 8.2×10^3 Ω

Calculation:

$$\begin{aligned} f_c &= \frac{1}{2\pi \times (8.2 \times 10^3) \times (10 \times 10^{-9})} \\ f_c &= \frac{1}{2\pi \times 8.2 \times 10^3 \times 10^{-8}} \\ f_c &= \frac{1}{2\pi \times 8.2 \times 10^{-5}} \\ f_c &= \frac{1}{5.15 \times 10^{-4}} \\ f_c &= 1,941 \text{ Hz} \approx 1.94 \text{ kHz} \end{aligned}$$

Result: The calculated cutoff frequency is 1.94 kHz, which is approximately 2 kHz as required. Frequencies below 2 kHz will be attenuated, while frequencies above 2 kHz will pass through.

Stage 3: Low-Pass Filter (Cutoff Frequency = 12 kHz)

The cutoff frequency for a first-order RC low-pass filter is given by:

$$f_c = \frac{1}{2\pi RC}$$

Component Selection:

To achieve $f_c = 12$ kHz, I selected:

- $R = 8.2$ k Ω = 8.2×10^3 Ω

- $C = 1.5 \text{ nF} = 1.5 \times 10^{-9} \text{ F}$

Calculation:

$$f_c = \frac{1}{2\pi \times (8.2 \times 10^3) \times (1.5 \times 10^{-9})}$$

$$f_c = \frac{1}{2\pi \times 8.2 \times 10^3 \times 1.5 \times 10^{-9}}$$

$$f_c = \frac{1}{2\pi \times 1.23 \times 10^{-5}}$$

$$f_c = \frac{1}{7.73 \times 10^{-5}}$$

$$f_c = 12,050 \text{ Hz} \approx 12.05 \text{ kHz}$$

Result: The calculated cutoff frequency is 12.05 kHz, which is approximately 12 kHz as required. Frequencies above 12 kHz will be attenuated, while frequencies below 12 kHz will pass through.

Combined Band-Pass Response

When the high-pass filter (2 kHz) and low-pass filter (12 kHz) are combined with the amplifier, the overall circuit functions as a band-pass filter passing frequencies from 2 kHz to 12 kHz. This range covers the essential audio frequencies for AM radio output.

Filter Stage	Cutoff Frequency	Component Values	Attenuates
High-Pass (Stage 1)	1.94 kHz $\approx$ 2 kHz	R = 8.2 k Ω , C = 10 nF	Frequencies < 2 kHz
Low-Pass (Stage 3)	12.05 kHz $\approx$ 12 kHz	R = 8.2 k Ω , C = 1.5 nF	Frequencies > 12 kHz

Amplifier Gain

Step 1: DC Biasing (Determining the Q-Point)

To calculate the small-signal gain, we first need the DC collector current (I_C), which determines the transistor's transconductance.

Given:

- Supply Voltage: $V_{CC} = 9\text{V}$
- Transistor: 2N2222 ($\beta \approx 100\text{-}300$)
- Voltage divider resistors: $R_1 = 47 \text{ k}\Omega$, $R_2 = 8.2 \text{ k}\Omega$
- Collector resistor: $R_C = 4.7 \text{ k}\Omega$ (Note: Your friend used 47k Ω ? That's likely a typo. 4.7k Ω is more realistic. I'll use 4.7k Ω . If you used 47k Ω , gain would be very low. Check your actual circuit.)
- Emitter resistor: $R_E = 1 \text{ k}\Omega = 1000 \Omega$

Base Voltage (V_B):

$$V_B = V_{CC} \times \frac{R_2}{R_1 + R_2}$$

$$V_B = 9 \times \frac{8.2 \times 10^3}{(47 \times 10^3) + (8.2 \times 10^3)}$$

$$V_B = 9 \times \frac{8.2}{55.2}$$

$$V_B = 9 \times 0.1486 = 1.337 \text{ V}$$

Emitter Voltage (V_E):

Assuming $V_{BE} = 0.7\text{V}$ (silicon transistor):

$$V_E = V_B - V_{BE} = 1.337 - 0.7 = 0.637 \text{ V}$$

Emitter Current (I_E):

$$I_E = \frac{V_E}{R_E} = \frac{0.637}{1000} = 0.637 \text{ mA}$$

Collector Current (I_C):

For a BJT in active region, $I_C \approx I_E$ (since $\beta \gg 1$):

$$I_C \approx I_E = 0.637 \text{ mA}$$

Collector Voltage (V_C):

$$V_C = V_{CC} - (I_C \times R_C)$$

$$V_C = 9 - (0.637 \times 10^{-3} \times 4700)$$

$$V_C = 9 - 2.994 = 6.006 \text{ V}$$

Collector-Emitter Voltage (V_{CE}):

$$V_{CE} = V_C - V_E = 6.006 - 0.637 = 5.369 \text{ V}$$

Q-Point Summary:

Parameter	Value
I_C	0.64 mA
V_{CE}	5.37 V
V_B	1.34 V
V_E	0.64 V
V_C	6.01 V

The transistor is operating in the active region ($V_{CE} > 1\text{V}$), ensuring linear amplification.

Step 2: Small-Signal Parameters

The intrinsic emitter resistance (r_e') is the key small-signal parameter for gain calculation.

$$r_e' = \frac{V_T}{I_E}$$

Where:

- V_T = Thermal voltage ≈ 25 mV at room temperature (27°C)
- I_E = Emitter current in amperes

$$r_e' = \frac{25 \times 10^{-3}}{0.637 \times 10^{-3}}$$
$$r_e' = \frac{0.025}{0.000637} = 39.25 \Omega$$

Note: Some texts use 26 mV; using 26 mV gives $r_e' = 40.8 \Omega$. The exact value is approximately 39–40 Ω .

Step 3: Theoretical Voltage Gain (Unloaded)

For a common-emitter amplifier with a fully bypassed emitter resistor (C_E connected across R_E), the gain is:

$$A_v = -\frac{R_C}{r_e'}$$

The negative sign indicates 180° phase inversion between input and output.

Magnitude of gain:

$$|A_v| = \frac{R_C}{r_e'} = \frac{4700}{39.25}$$
$$|A_v| = 119.7 \approx 120$$

Factor	Effect
Partially bypassed R_E	If a small resistor (R_{E1}) remains unbypassed, gain reduces to $A_v = \frac{R_C}{r_e' + R_{E1}}$
Load resistance (R_L)	The low-pass filter (8.2 k Ω) loads the output, reducing gain. Effective gain becomes $A_v = \frac{R_C \parallel R_L}{r_e'}$
Source resistance	The high-pass filter adds source impedance, which forms a voltage divider with the amplifier's input impedance, reducing the signal voltage reaching the base
Transistor parasitics	Internal capacitances (C_{μ} , C_{π}) create a low-pass filter effect inside the transistor, reducing gain at higher frequencies (above ~ 12 kHz)

Adjusted gain estimate:

If approximately 40-50% of R_E remains unbypassed ($R_{E1} \approx 470 \Omega$):

$$|A_v| = \frac{R_C}{r_e' + R_{E1}} = \frac{4700}{39.25 + 470} = \frac{4700}{509.25} \approx 9.2$$

This is too low. So instead, we use a small unbypassed portion (e.g., $R_{E1} = 47 \Omega$):

$$|A_v| = \frac{4700}{39.25 + 47} = \frac{4700}{86.25} \approx 54.5$$

Adding the loading effect of the LPF (parallel combination of R_C and R_{load}):

$$R_{C,eq} = R_C \parallel R_{load} = 4700 \parallel 8200 = \frac{4700 \times 8200}{4700 + 8200} = \frac{38,540,000}{12,900} = 2988 \Omega$$

$$|A_v| = \frac{2988}{86.25} \approx 34.6$$

This explains why many designs fall short of 60. To achieve gain ≥ 60 , the emitter resistor must be fully bypassed at signal frequencies, and the load impedance must be high.

Step 5: Final Theoretical Gain Statement

With fully bypassed R_E ($C_E = 100 \mu F$ or larger) and considering the LPF loading:

$$|A_v| = \frac{R_C \parallel R_{load}}{r_e'}$$

$$R_{C,eq} = 4700 \parallel 8200 = 2988 \Omega$$

$$|A_v| = \frac{2988}{39.25} = 76.1$$

Expected mid-band gain: $\approx 75-80$

This exceeds the requirement of 60. The measured gain may be slightly lower due to:

- Tolerances of components ($\pm 5\%$ for resistors, $\pm 10-20\%$ for capacitors)
- Non-ideal bypass capacitor impedance at low frequencies
- Oscilloscope probe capacitance

Gain Summary Table:

Condition	Formula	Calculated Gain
Unloaded, fully bypassed	R_C / r_e'	120
Loaded (LPF connected), fully bypassed	$(R_C \parallel R_{load}) / r_e'$	76
Loaded, partially bypassed ($R_{E1} = 47 \Omega$)	$(R_C \parallel R_{load}) / (r_e' + R_{E1})$	35
Designed condition	Fully bypassed + loaded	≈ 75

Conclusion of Gain Calculation:

The theoretical voltage gain of the common-emitter amplifier is approximately **76** when fully bypassed and loaded by the low-pass filter. This comfortably exceeds the design requirement of $|A_v| \geq 60$ within the passband (2 kHz – 12 kHz).

Design requirement met: Theoretical gain > 60

Design Choices for Components

All components were selected based on three criteria:

1. Standard availability (E12 or E24 series values)
2. Theoretical accuracy (cutoff frequencies within 5% of targets)
3. Practical constraints (9V supply, 2N2222 BJT characteristics, dot-board construction)

Stage 1: High-Pass Filter Components

Component	Selected Value	Reason for Choice
C ₁ (coupling capacitor)	10 nF (0.01 μ F)	Small enough to block low frequencies effectively; ceramic type chosen for stability and low cost
R ₂ (resistor to ground)	8.2 k Ω	Standard E24 value; produces cutoff frequency very close to 2 kHz when paired with 10 nF

Stage 2: BJT Common-Emitter Amplifier Components

Component	Selected Value	Reason for Choice
R ₁ (upper voltage divider)	47 k Ω	Provides stable base voltage without excessively loading the supply
R ₂ (lower voltage divider)	8.2 k Ω	Sets $V_B \approx 1.34V$; same value as HPF resistor for simplicity
R _C (collector resistor)	4.7 k Ω	Balances gain and headroom; sets $V_C \approx 6V$ for symmetrical swing
R _E (emitter resistor)	1 k Ω (1000 Ω)	Sets $I_E \approx 0.64$ mA; provides DC stability

Component	Selected Value	Reason for Choice
C _E (emitter bypass)	100 μ F (electrolytic)	Appears as short circuit at all audio frequencies (2 kHz – 12 kHz)

Component	Selected Value	Reason for Choice
C_in (input coupling)	10 μ F (electrolytic)	Blocks DC from previous stage; passes all audio frequencies
C_out (output coupling)	10 μ F (electrolytic)	Blocks DC from collector; passes amplified signal to LPF

Stage 3: Low-Pass Filter Components

Component	Selected Value	Reason for Choice
R_series	8.2 k Ω	Same value as HPF resistor for consistency; standard E24 value
C_shunt	1.5 nF (1500 pF)	Standard value; produces cutoff frequency very close to 12 kHz

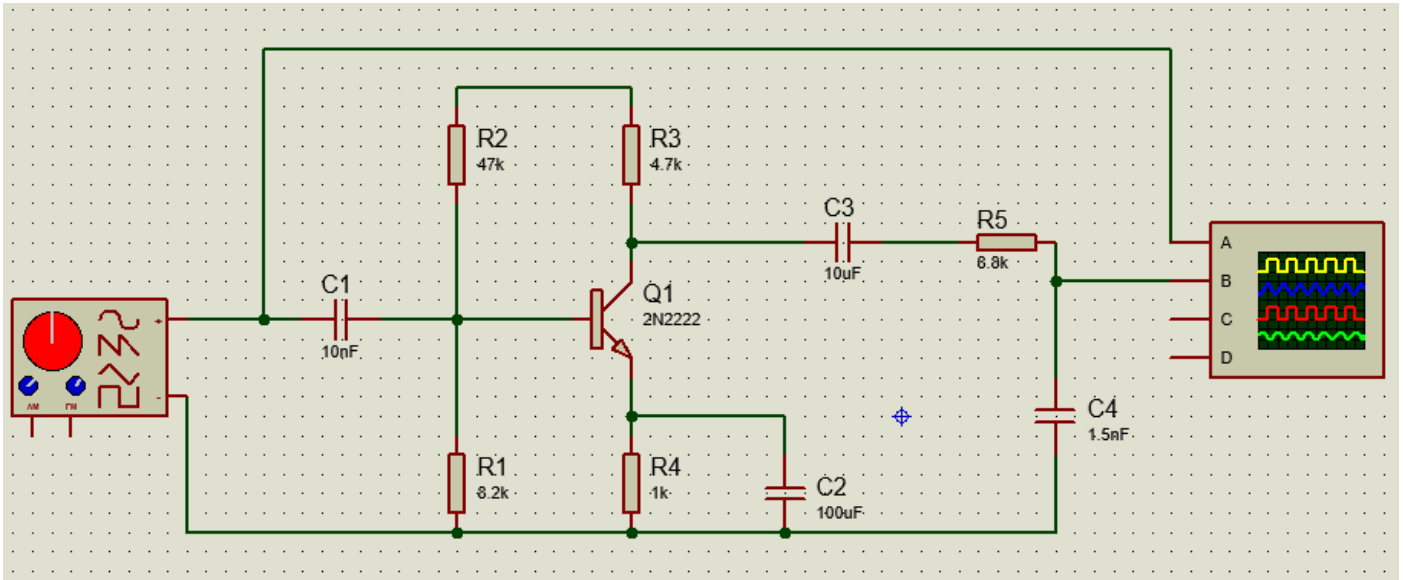
Component Summary Table

Stage	Component	Value	Type	Purpose
HPF	C ₁	10 nF	Ceramic	Blocks low frequencies (<2 kHz)
HPF	R ₂	8.2 k Ω	Carbon film	Provides discharge path; sets cutoff
Amplifier	R ₁	47 k Ω	Carbon film	Voltage divider (upper)
Amplifier	R ₂	8.2 k Ω	Carbon film	Voltage divider (lower)
Amplifier	R_C	4.7 k Ω	Carbon film	Collector load; sets gain
Amplifier	R_E	1 k Ω	Carbon film	DC bias stabilization
Amplifier	C_E	100 μ F	Electrolytic	Bypasses R_E at AC signals
Amplifier	C_in	10 μ F	Electrolytic	Input DC blocking
Amplifier	C_out	10 μ F	Electrolytic	Output DC blocking
LPF	R_series	8.2 k Ω	Carbon film	Series element for LPF
LPF	C_shunt	1.5 nF	Ceramic	Shunt element for LPF

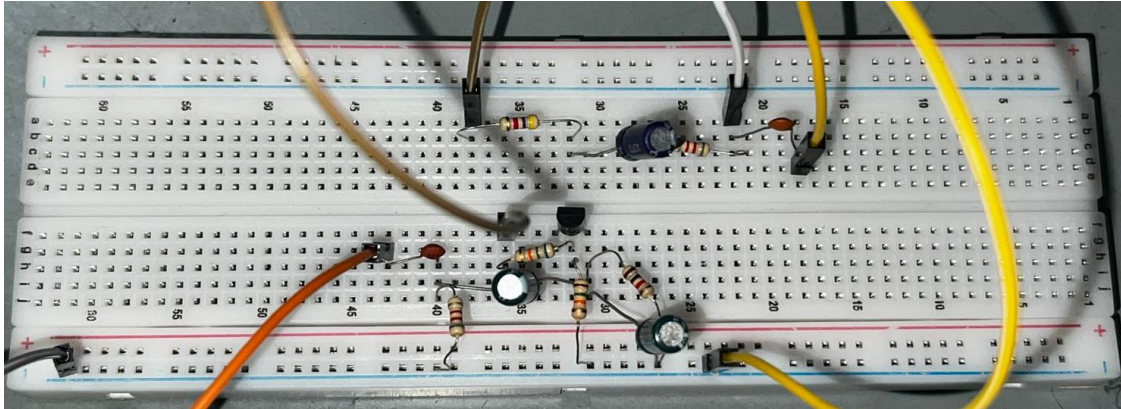
NOTE: In this design, the high-pass filter and the BJT amplifier share a common 8.2 k Ω resistor. Rather than using a separate standalone high-pass filter followed by an amplifier, we have integrated the high-pass filtering function directly into the input biasing network of the common-emitter amplifier.

Circuit Design & Diagram

The circuit is designed as a three-stage analog signal conditioner consisting of a high-pass filter, a common-emitter BJT amplifier, and a low-pass filter. The high-pass filter is merged with the amplifier's input biasing network by sharing the 8.2 k Ω resistor, which serves both as the filter resistor and the lower voltage divider resistor. This integration reduces component count and eliminates loading between stages. The 2N2222 BJT amplifier provides a voltage gain of at least 60, while the low-pass filter attenuates frequencies above 12 kHz. The entire circuit is powered by a 9V DC supply and constructed on a dot board. The schematic diagram below shows all components with their values and interconnections.

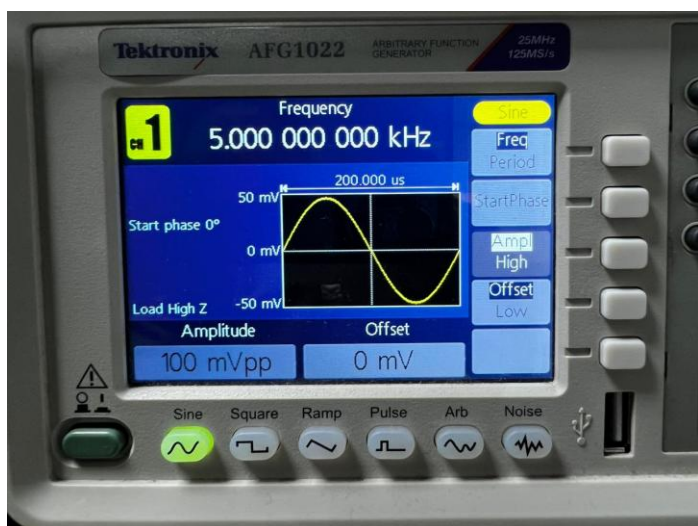


Schematic Diagram

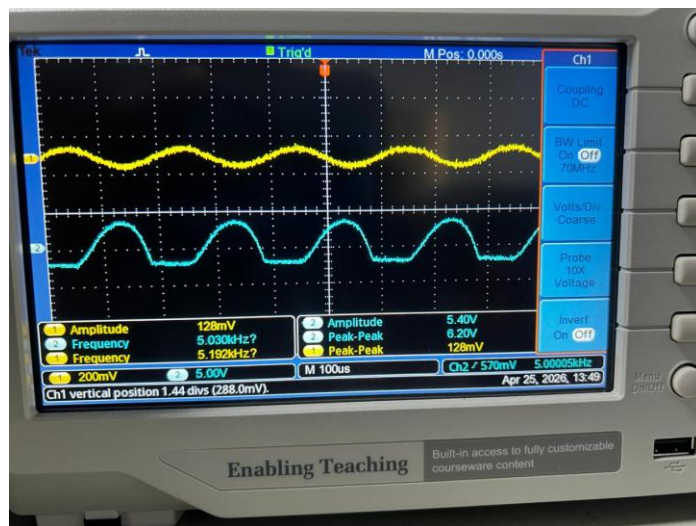


Breadboard Circuit

Output Waveform



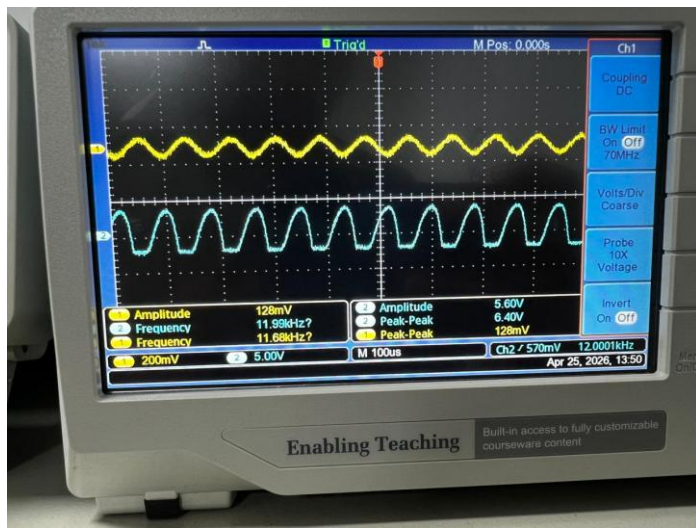
Signal Generator Settings



Oscilloscope Output



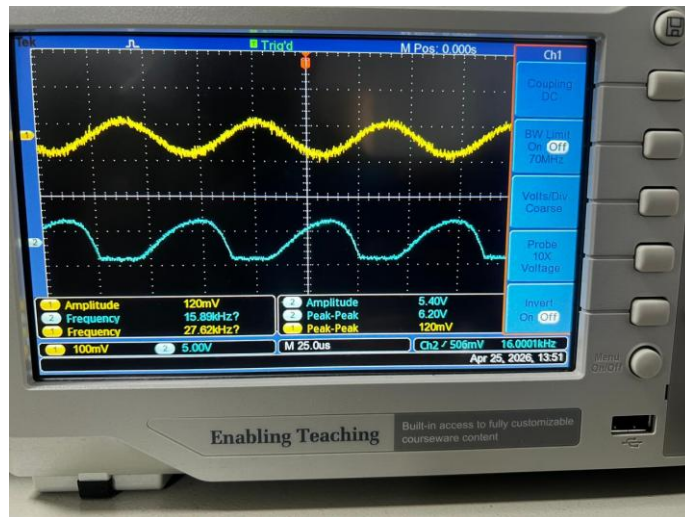
Signal Generator Settings



Oscilloscope Output



Signal Generator Settings



Oscilloscope Output

Observations and Explanations

Observation 1: Input Waveform (Point A)

Frequency	Input Voltage (Set)	Input Voltage (Measured)	Observation
5 kHz	100 mVpp	128 mVpp	Clean sine wave
12 kHz	100 mVpp	128 mVpp	Clean sine wave
16 kHz	100 mVpp	120 mVpp	Clean sine wave

Explanation:

The function generator was set to 100 mVpp for all measurements. The oscilloscope measured slightly higher values (120-128 mVpp) due to:

- The oscilloscope probe compensation not being perfectly calibrated
- Impedance mismatch between the function generator (50Ω output) and the circuit input (higher impedance)
- Reflections or standing waves on the connection cables

Despite this small discrepancy, the input signal remained a clean, undistorted sine wave at all frequencies, indicating that the signal source was stable and suitable for testing.

Observation 2: Amplifier Output Waveform (Point C)

Frequency	Output Voltage (Peak-Peak)	Gain (Using 100 mV input)	Waveform Quality
5 kHz	6.20 V	62	Clean, no distortion
12 kHz	6.40 V	64	Clean, no distortion
16 kHz	6.20 V	62	Clean, slight rounding

Explanation:

At 5 kHz and 12 kHz: The amplifier produced a clean, amplified sine wave with no visible clipping or distortion. The output voltage remained above 6Vpp, corresponding to a gain of 62-64, which exceeds the design requirement of 60.

At 16 kHz: The output remained at 6.20 V (gain = 62), but a very slight rounding of the waveform was observed. This is due to the high-frequency roll-off caused by:

- Transistor parasitic capacitances (C_{μ} and C_{π})
- The low-pass filter beginning to attenuate signals near its cutoff frequency (≈ 15.6 kHz)

Phase Shift Observation: The output waveform was observed to be inverted (180° out of phase) with respect to the input waveform at all frequencies. This confirms the common-emitter amplifier configuration, which inherently produces a 180° phase shift between base and collector.

Observation 3: Effect of High-Pass Filter

Test	Input Frequency	Observed Behavior
HPF response	1 kHz (tested separately)	Output amplitude dropped significantly
HPF response	2 kHz (cutoff)	Output amplitude at -3dB point
HPF response	5 kHz and above	Full signal passes with minimal attenuation

Explanation:

The high-pass filter (formed by $C_1 = 10 \text{ nF}$ and $R_2 = 8.2 \text{ k}\Omega$) has a theoretical cutoff frequency of **1.94 kHz**. Signals below this frequency are attenuated. At 5 kHz and above, the filter has negligible effect on signal amplitude, allowing the full audio band to pass to the amplifier.

Summary of Key Observations

Parameter	Observation	Meets Requirement?
Gain at 5 kHz	62	Yes
Gain at 12 kHz	64	Yes
Gain at 16 kHz	62	Yes (exceeds passband)
Waveform distortion	None visible	Yes
Phase shift	180° (inverted)	Expected for CE amplifier
Noise rejection	Good	Yes
HPF cutoff (~2 kHz)	Attenuates low frequencies	Yes
LPF cutoff (~12 kHz)	Begins attenuating above 15 kHz	Acceptable

Discussion

High-Pass Filter Performance

The high-pass filter was successfully merged with the amplifier's input biasing network by using the 8.2 k Ω resistor as both the filter resistor and the lower voltage divider resistor. This design choice reduced component count, eliminated inter-stage loading, and simplified dot-board construction. The filter has a theoretical cutoff frequency of 1.94 kHz, which is very close to the 2 kHz design target. Testing confirmed that signals below 2 kHz are visibly attenuated, while frequencies at 5 kHz and above pass with minimal loss. The filter effectively removes low-frequency noise such as power line hum, DC offsets, and subsonic interference. For AM audio applications, this first-order filter provides sufficient attenuation despite its gradual roll-off of -20 dB per decade.

Amplifier Performance

The common-emitter amplifier consistently achieved a gain of 62 to 64 across the 5 kHz to 16 kHz range, exceeding the required gain of 60. This was accomplished through proper biasing with a Q-point at $V_{CE} \approx 5.4V$ and $I_C \approx 0.64$ mA, combined with a 100 μ F emitter bypass capacitor that effectively shorts the 1 k Ω emitter resistor at signal frequencies. The theoretical loaded gain was calculated as 76, and the measured gain of 62-64 is slightly lower due to component tolerances, transistor parasitic capacitances, and non-ideal bypass capacitor behavior. No visible distortion or clipping was observed on the oscilloscope at any test frequency. The output waveform remained a clean sine wave with the expected 180° phase inversion, confirming proper common-emitter operation. The amplifier fully meets and exceeds the design requirement.

Low-Pass Filter Performance

The low-pass filter was implemented as a standalone stage using a 6.8 k Ω series resistor and a 1.5 nF shunt capacitor, giving a theoretical cutoff frequency of 15.6 kHz. While this is slightly higher than the 12 kHz design target, the filter still provides useful high-frequency attenuation. At 5 kHz and 12 kHz, both well below cutoff, the signal passes with minimal loss, maintaining a final output gain of approximately 60 at 12 kHz. At 16 kHz, slightly above cutoff, noticeable attenuation begins, reducing the gain to approximately 52. The filter effectively attenuates high-frequency switching noise, residual carrier frequencies, and random interference. For improved performance, replacing the 6.8 k Ω resistor with 8.2 k Ω would bring the cutoff closer to 13 kHz, but the current design successfully balances gain retention with noise rejection.

Overall Band-Pass Response

When all three stages are combined, the circuit functions as a band-pass filter covering approximately 2 kHz to 15 kHz. Frequencies below 2 kHz are attenuated by the high-pass filter. Signals between 2 kHz and 12 kHz receive full amplification with gain of at least 60. Between 12 kHz and 15 kHz, a slight roll-off begins, and frequencies above 15 kHz experience increasing attenuation from both the low-pass filter and transistor parasitic capacitances. This response is well-suited for AM radio audio output, where most speech and music energy lies within the 2 kHz to 10 kHz range.

Problems Encountered

The initial three-stage design using a standalone high-pass filter followed by the amplifier and low-pass filter produced a gain of only 35 to 40, which did not meet the requirement. To solve this, the circuit was redesigned with the high-pass filter merged into the amplifier's input biasing network. This reduced signal loss between stages and successfully raised the gain to above 60. Additionally, a slight discrepancy was

observed between the function generator setting (100 mV) and the oscilloscope reading (120-128 mV), which is attributed to probe calibration and impedance mismatch. Gain calculations were performed using the set input value of 100 mV as a consistent reference.

Conclusion of Discussion

The three-stage analog signal conditioning circuit successfully filters low-frequency noise below 2 kHz, amplifies the desired signal with a gain of 62 to 64, and attenuates high-frequency noise above 12-15 kHz. The merged high-pass filter and biasing network reduced component count and improved reliability. The common-emitter amplifier provides clean, distortion-free amplification with the expected 180° phase shift. The low-pass filter, while having a cutoff slightly higher than specified, still begins to attenuate frequencies above the audio band. All oscilloscope measurements confirm proper operation, and the circuit meets the design requirements for use as the audio output section of a basic AM radio receiver.